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# COMPSCI 602

## Project Report 4

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Anonymous

**Project Title:** Tracing the Positional Decay of the Refusal Direction in Safety-Aligned LLMs under Prefilling Attacks

### 1 Introduction

**System.** The system under study is Llama-3.1-8B-Instruct [4], a safety-aligned large language model with 8 billion parameters. This model exhibits the shallow safety alignment phenomenon: safety constraints are concentrated in the first few generated tokens and decay as generation proceeds [5].

**Task.** The task is safety-constrained response generation. Given a harmful query (e.g., requesting instructions for dangerous activities), the model should produce a refusal. We study the transition from refusal to compliance as the model generates tokens beyond the initial refusal prefix.

**Environment.** The environment consists of adversarial prompt sets that reliably trigger shallow alignment failures. We primarily study *prefilling attacks*, where the model is forced to begin generation with a compliant prefix [5]. As a secondary comparison, we may also examine *adversarial suffix attacks* [8] if time permits. Harmful prompts are drawn from AdvBench [8] and JailbreakBench [2], with benign prompts from Alpaca [6] as controls.

**Phenomena.** The primary phenomenon is the *positional decay of a safety-relevant internal representation* during autoregressive generation. Prior work has established a sparse “refusal direction” in the residual stream that mediates refusal behavior [1]. Other studies have proposed additional candidate substrates, such as specific “safety neurons” [3] and individual “safety heads” in the attention mechanism [7], but the temporal dynamics of these representations across token positions are not well understood. We focus on the refusal direction as our primary mechanism and aim to identify causal contributors to its decay under prefilling attacks.

### 2 Research Question 1: How Does the Safety Signal Change?

**Research Question 1:** *How does the refusal-direction signal change across token positions and layers under prefilling attacks?*

This question is *mechanistic*: we seek not merely to observe that decay occurs, but to understand how the refusal-direction representation evolves as the model generates tokens, and how this evolution varies across layers. While Qi et al. [5] demonstrated behavioral decay and Ardit et al. [1] identified the refusal direction at pre-generation positions, no prior work has traced the positional dynamics of this representation through the model’s layers during generation. Characterizing how the safety signal weakens is a prerequisite for deeper explanations and for developing robust defenses.

**Hypothesis 1a (Decay Exists and Is Measurable):** The cosine similarity between the residual stream activations and the refusal direction will show a statistically significant negative trend as token position increases during generation under prefilling attacks. This would confirm that the behavioral phenomenon observed by Qi et al. has a measurable internal correlate.

*Falsification criterion:* If the refusal direction magnitude remains stable (no statistically significant change) across token positions, or increases at later positions, this hypothesis is falsified.

**Hypothesis 1b (Decay Propagates Through Mid-to-Late Layers):** The decay will not occur uniformly across layers. Instead, mid-to-late layers (layers 15–25 in the 32-layer model) will show faster decay than early layers, suggesting that safety information persists in early representations but is progressively overwritten as information flows through the network.

*Falsification criterion:* If all layers show uniform decay rates, or if early layers show faster decay than late layers, this hypothesis is falsified.

### 3 Research Question 2: How Can Causal Intervention Restore Safety?

**Research Question 2:** *How does patching safety-relevant activations from early token positions into later positions affect the model’s behavior, and what does this reveal about the mechanism of decay?*

This question is *causal*: we directly manipulate internal representations to test whether the decay is caused by (a) a *loss* of safety-relevant information, or (b) *active suppression* by other model components. If early-position activations can restore refusal when patched into later positions, this implies that the information was lost rather than overridden. If patching fails, it suggests that later-position computations actively suppress the safety signal.

**Hypothesis 2a (Restoration Is Possible):** Patching the refusal direction component of residual stream activations from token position  $t = 1$  into positions  $t \geq 5$  will produce a statistically significant increase in refusal rate compared to the unpatched (attacked) baseline.

*Falsification criterion:* If patching early activations produces no statistically significant increase in refusal rate, this hypothesis is falsified.

**Hypothesis 2b (Restoration Requires Correct Layer Targeting):** The effectiveness of restoration will depend on *where* in the network we patch. Restoration will be most effective when patching at mid-to-late layers (where the refusal direction is strongest), rather than at early layers. This would suggest that the decay mechanism operates at specific points in the forward pass.

*Falsification criterion:* If early-layer patching is equally or more effective than mid-to-late layer patching, this hypothesis is falsified.

### 4 Research Question 3: Do Different Attacks Work Through the Same Mechanism? (Exploratory)

**Research Question 3:** *Through what distinct or shared internal mechanisms do prefilling attacks and adversarial suffix attacks suppress the safety signal?*

This question probes whether the decay phenomenon has a *single underlying cause* or whether different attacks exploit different vulnerabilities. Prefilling attacks force compliant tokens into the generation context, while adversarial suffixes operate by hijacking attention [1]. If both converge on the same internal decay pattern, this suggests a fundamental architectural vulnerability. If they diverge, each may require distinct defenses. This research question is exploratory and will be pursued if time permits after addressing RQ1 and RQ2.

**Hypothesis 3a (Shared Vulnerability):** Both attack types will produce qualitatively similar decay patterns in the refusal direction magnitude across token positions, suggesting they exploit a shared vulnerability in the model’s safety mechanism.

*Falsification criterion:* If the decay curves show qualitatively different shapes (e.g., one monotonically decreasing, one non-monotonic), this hypothesis is falsified.

**Hypothesis 3b (Distinct Pathways):** The two attack types will affect different layers preferentially. Prefilling attacks will primarily suppress the refusal direction in early-to-mid layers (where context is integrated), while adversarial suffix attacks will primarily affect mid-to-late layers (where attention is hijacked).

*Falsification criterion:* If both attack types show maximum decay at the same layers, this hypothesis is falsified.

Note: Hypotheses 3a and 3b represent competing predictions. Our experiments will determine which (if either) is supported by the evidence.

## References

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